

FUNDAMENTAL OF ELECTRONICS

ET 8213

Course Information and Overview

Instructor : Mr. William Moshi
Office : CoICT, ETE Dept R3
Phone: : +255 762473864
Email: :willymosh@gmail.com

Course Objectives and Outcome

- ❑ To describe Operational Amplifiers and its characteristics
- ❑ To construct single stage amplifiers
- ❑ To describe and analyze the operation of several types of op-amp application circuits
- ❑ Describe filters operations and characteristics
- ❑ Types of oscillators RC, LC feedback oscillator and operation principles
- ❑ Design and describe voltage regulators

Course Information

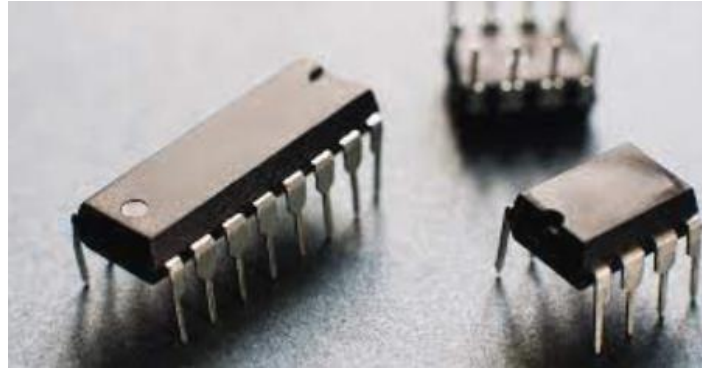
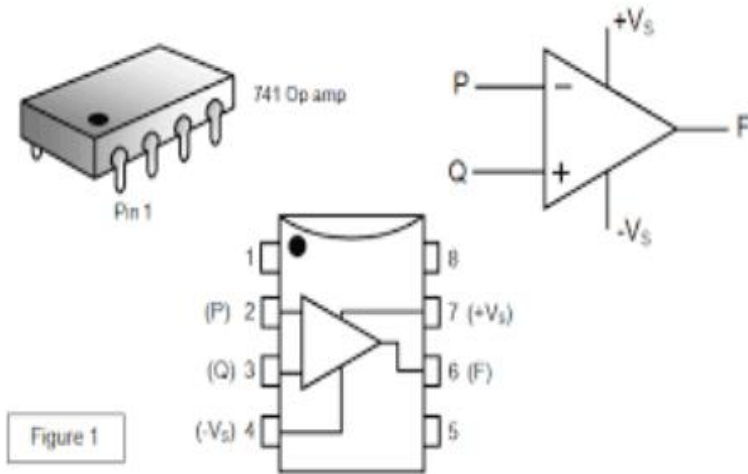
- ❑ Delivery mode : 2 hrs lecture
- ❑ Schedule : Tue LBPR01, Lecture/tutorial
Time: 16:50 - 19:05
- ❑ Course assessment: Assignment, class attendance & Test 40%
-University Exam 60%

Course Content

► Operational amplifier:

Introduction to op-amp, input modes, negative feedback, open loop and closed loop operation, Basics of differential amplifier, Ideal op-amp, inverting op-amp, non-inverting op-amp, op-amp parameters (frequency response, gain-bandwidth, slew rate, offset currents and voltages, PSRR). Op-amp applications: summing amplifier, voltage subtraction, integration, differentiation, voltage comparators (hysteresis and Schmitt triggers, and applications of voltage comparators).

Course Content....

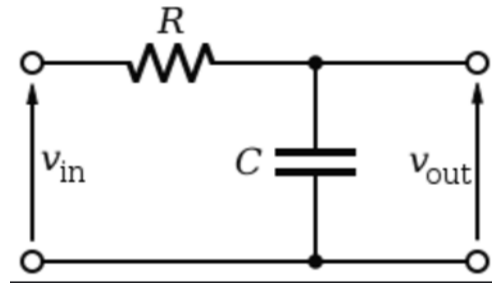


- Operational Amplifier symbol and circuit diagram

Course Content....

❑ Filters:

Basic filter response, characteristics, Low pass, high pass, band pass, band stop and their applications

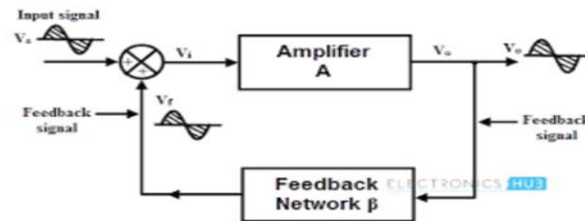


Simple filter and its circuit diagram

Course Content....

❑ Oscillators :

Operation, Phase shift Oscillator, Wien bridge Oscillator, Tuned Oscillator circuits, Crystal Oscillator. (BJT Version Only) Simple design methods of Oscillators.

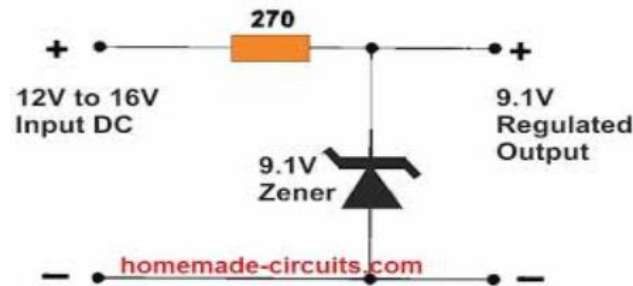
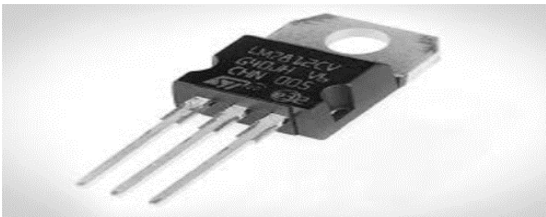


Oscillator simple circuit diagram

Course Content....

❑ Regulators:

Describe voltage regulation, Line regulation and Load regulation ,Basic Liner shunt and Linear series regulators , Basic switching regulator, IC voltage regulators.



Regulator circuit diagram

Recommended References/Textbooks

- ❑ C.A. Schuler, —Electronics Principles and ApplicationsII, Fifth edition, (1999), Glencoe McGraw-Hill
- ❑ V.K. Mehta & S. Mehta, —Principles of _ElectronicsII, (2004), S. Chand & Company.
- ❑ Floyd, —Electronics DevicesII, Fifth Edition, (20010, Saurabh Print, India
- ❑ Ian Sinclair,IIPranctical Electronics HandbookII, Fourth Edition, (1999)
- ❑ Tom Duncun, Success in ElectronicsII, 2nd Edition (1997), John Murray Publishers.
- ❑ A.R. Hambley, ElectronicsII, Second Edition, (2002), Tom Robbins Publishers.